Test report 11 – colors, multi-filament 3MF, dark theme

cad-eval (palette), cad-engine (colors, color_export), cad-session, cad-cli, cad-render, web (color list, theme) · 2026-10-23 · Branch worktree-agent-a2ab30636d524862b · Commit 23ec721 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

cargo test --workspace	clippy · fmt	Vitest · Playwright	Tauri-E2E	3MF checked independently	OpenSCAD · STEP
388 / 388	**clean · clean**	**193 · 84 / 84**	**14 / 14**	**passed**	**33/33 · 48/48**
of which 31 new color tests	-D warnings	of which 7 new scenarios (F01–F07)	tauri-driver + WebKitWebDriver, Xvfb	lib3mf, trimesh, Bambu model, PrusaSlicer	compare.py · verify.py

Palette: smallest ΔE_{OK}
0.107 · 0.070
normal · deuteranopia

1 Task and result

Three user requests: **multiple colors** with an export that Bambu Studio prints "fairly easily with several filaments"; replace the CSS colors with a **small, good-looking preset** (English names only, muted tones, warning and random color for unknown names); a thoroughly reworked **dark theme**. Implemented:

- **Palette** with 16 colors in `cad-eval` (one source, generated for the UI), unknown names → warning „Farbe `x` unbekannt – Zufallsfarbe“ ("color x unknown – random color") and a stable color from the hash of the name; hex, vectors and `alpha` exact (section 2).
- **Color regions**: each part is split via kernel booleans into bodies per color (cached), sum of the volumes = part volume; **filament slots** from `color("teal", filament = 2)` or automatic (section 3).
- **3MF for Bambu Studio / OrcaSlicer**: one object per part, one sub-part per color at the same origin, color group and `Metadata/model_settings.config` with `extruder` per sub-part – the format worked out from the slicers' source code (section 5), checked with lib3mf, trimesh, a model of the Bambu importer and PrusaSlicer (section 6). In addition "one STL per color" as a ZIP. CLI: `cad export -o x.3mf` is multi-color (`--single-color`), `cad outline` and `cad measure --colors` show colors, slots, volume and mass.
- **UI**: "Colors & filaments" (German UI: „Farben & Filamente“) (swatch, name, lines, slot selector that writes `filament = n` into the code, volume and mass for PLA 1.24 g/cm³), color per face in the view, "Show color regions" (German UI: „Farbregionen zeigen“) (section 7).
- **Dark theme**: choice System (default, follows `prefers-color-scheme`, also in Tauri)/Light/Dark, token set with WCAG check (text ≥ 4.5:1), editor components with their own default theme adapted, palette colors slightly lightened in the dark theme; `cad render --theme dark` (section 8).

2 Palette

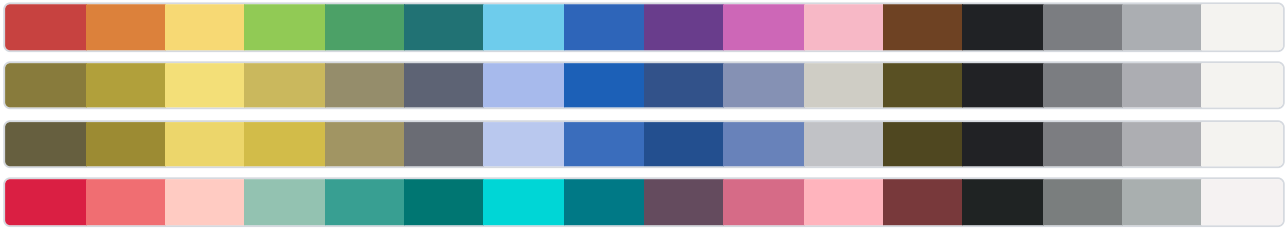
The names chosen are those that occur most often in OpenSCAD files – in the corpus of OpenSCAD examples and libraries in the container red (11), green (7), cyan (7), yellow, gray, blue, black (2 each), magenta, lime; from experience with widely used projects additionally white, orange, silver, purple – plus teal, pink, brown as common filament colors. The proposal `indigo`/`slate` was dropped: indigo lies between blue and purple (hard to separate with deuteranopia), slate between gray and blue; `silver` and `magenta` are more frequent in OpenSCAD files. `grey`, `aqua`, `fuchsia` are the CSS spellings of gray, cyan, magenta; there are no German names.

Values: target values in OKLCH (lightness L, chroma C ≤ 0.18 – muted, no pure RGB corners –, hue h), optimized by a search for the smallest distance in OKLab for normal vision and for simulated deuteranopia and protanopia (Machado, Oliveira, Fernandes 2009, severity 1.0, in linear sRGB). The formulas are in `palette.rs`; `cargo test -p cad-eval --test palette` enforces the minimum distances, `tools/palette/palette.py` recomputes the same values.

Name	Base value (export)	shown light on #e7eaee	shown dark on #16191d	Deut.	Prot.	Trit.	OKLCH (L · C · h)

red	#c74240	#c74240 4.06:1	#c74240 3.60:1				0.568 · 0.170 · 25.1
orange	#dc813b	#dc813b 2.40:1	#dc813b 6.08:1				0.690 · 0.140 · 55.1
yellow	#f7d974	#f7d974 1.15:1	#f7d974 12.71:1				0.890 · 0.125 · 93.2
lime	#91ca55	#91ca55 1.61:1	#91ca55 9.05:1				0.775 · 0.160 · 131.0
green	#4ca167	#4ca167 2.64:1	#4ca167 5.54:1				0.641 · 0.120 · 151.9
teal	#217274	#217274 4.67:1	#307e80 3.71:1				0.506 · 0.075 · 197.5
cyan	#6eccec	#6eccec 1.51:1	#6eccec 9.67:1				0.799 · 0.100 · 222.5
blue	#2e65b9	#2e65b9 4.73:1	#3a71c6 3.66:1				0.515 · 0.145 · 258.8
purple	#693d8c	#693d8c 6.57:1	#754999 2.66:1				0.450 · 0.130 · 307.7
magenta	#cd67b7	#cd67b7 2.80:1	#cd67b7 5.22:1				0.660 · 0.161 · 336.1
pink	#f7b8c6	#f7b8c6 1.38:1	#f7b8c6 10.60:1				0.844 · 0.075 · 4.1
brown	#6e4223	#6e4223 7.05:1	#7a4d2e 2.46:1				0.425 · 0.076 · 54.3
black	#202225	#202225 13.21:1	#3b3d41 1.62:1				0.251 · 0.006 · 258.4
gray	#7b7d81	#7b7d81 3.42:1	#7b7d81 4.28:1				0.589 · 0.007 · 264.5
silver	#abaeb2	#abaeb2 1.85:1	#abaeb2 7.92:1				0.750 · 0.007 · 255.5
white	#f4f3f0	#e9e8e5 1.02:1	#f4f3f0 15.89:1				0.964 · 0.004 · 91.4
Default (without color())	#c3c8cf	Material without `color()` in exports and in the color list; in the view the model color of the theme (`--view-model`)					

Strips: palette normal, simulated deuteranopia, protanopia, tritanopia.



Vision	smallest distances ΔE_{OK} (OKLab, ~0.02 just visible)
normal vision	cyan/silver **0.107** · teal/gray **0.111** · pink/silver **0.122** · teal/blue **0.128** · blue/purple **0.132**
deuteranopia (green)	red/green **0.070** · green/gray **0.076** · blue/purple **0.077** · orange/lime **0.078** · cyan/silver **0.080**
protanopia (red)	orange/green **0.052** · cyan/pink **0.054** · teal/gray **0.058** · pink/silver **0.063** · red/teal **0.078**
tritanopia (blue)	teal/blue **0.031** · yellow/pink **0.056** · lime/silver **0.056** · orange/magenta **0.058** · purple/brown **0.067**

- **Display per theme** (same names, same hue): in the dark theme black is raised to L 0.36 (#3b3d41) and dark tones (teal, blue, purple, brown) +0.04 lighter; in the light theme white is lowered to L 0.93 (#e9e8e5). Exports, STEP and 3MF use the base values. The contrast given in the table is that of the flat color against the background of the 3D view; bodies are shaded and have edge lines, so white (light) and black (dark) are also visible, but only barely.
- **Color vision deficiency:** 16 colors cannot be separated well for all forms. With deuteranopia red/green (0.070), green/gray and blue/purple are closest, with protanopia orange/green (0.052), with the rare tritanopia teal/blue (0.031) – the list therefore always shows name and slot next to the color.
- **Unknown names:** hue from FNV-1a 64 of the lower-cased name, L = 0.65 and C = 0.125 (medians of the twelve chromatic palette colors, C reduced until the value lies in sRGB). Same name → same color, also across sessions, CLI and UI.

Input	Color	Source
`color("lightblue")`	#bd70af	random
`color("Rot")`	#d06f65	random
`color("salmon")`	#8a82d7	random
`color("navy")`	#b78520	random

`color("gold")`	#219bce	random
`color("olive")`	#cf6c7f	random

```
zufall.scad:1:7: Warnung (Auswertung): Farbe `lightblue` unbekannt – Zufallsfarbe (Palette: red, orange,
yellow, lime, green, teal, cyan, blue, purple, magenta, pink, brown, black, gray, silver, white)
zufall.scad:2:7: Warnung (Auswertung): Farbe `Rot` unbekannt – Zufallsfarbe (Palette: red, orange, yellow,
lime, green, teal, cyan, blue, purple, magenta, pink, brown, black, gray, silver, white)
zufall.scad:3:7: Warnung (Auswertung): Farbe `salmon` unbekannt – Zufallsfarbe (Palette: red, orange, yellow,
lime, green, teal, cyan, blue, purple, magenta, pink, brown, black, gray, silver, white)
```

3 Rules: color regions and filaments

Color regions (volume per color)

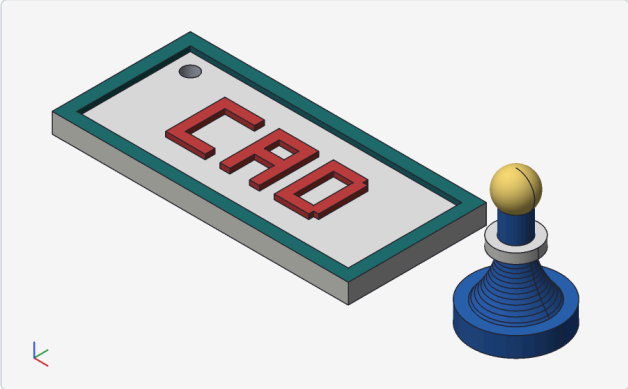
- The body of a colored subtree belongs to its color: the result of the `color()` node, placed in the coordinate system of the part (local matrices and `resize` factors along the path).
- If colors overlap, the **later statement** wins, the earlier one is reduced by it; "later" means pre-order in the tree: an inner `color()` after the outer one, a later sibling after the earlier one.
- **Subtracted material applies to all**: each piece is intersected with the result (`difference` removes the hole from all colors); colors that occur only in subtracted operands get no slot.
- Material in no colored subtree (e.g. the fillet between two colors) belongs to the color on the part, else to the default entry (if the part has uncolored primitives), else to the first color of the part.
- Computation: `Stück(C) = R ∩ M(C) − ⋃ M(später)` (piece of C = result ∩ material of C − union of later materials), remainder `R − ⋃ M`, pieces of one color united, empty ones dropped; about three booleans per color node. Result per part is a list (color, slot, body) with volume; if the sum differs from the part volume, the engine warns („Farbregionen ungenau“, "color regions inaccurate"). Cached by part, color nodes, placements and colors; color and slot are **not** part of the node keys – a new color or a new slot recomputes no node (test `regions_are_cached…`). Without color nodes below the part there is no boolean: one region from the bodies of the part, volume = cached part volume (test `uncoloured_part_needs_no_boolean`).

Filament slots

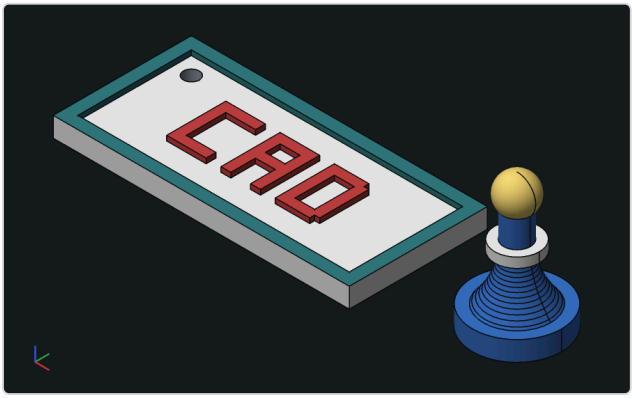
- `color("teal", filament = 2)` sets the slot of the color (1–16, named argument; the first specification applies, a contradiction warns).
- Material without `color()` ("default", #c3c8cf) prints with slot 1.
- All other colors get the lowest free slot in the order of their first occurrence. The same color in another spelling (`"gray"`, `"GREY"`, `"#7b7d81"`) is the same entry.
- **Changing the slot in the UI = text edit** (`SetFilament`): every `color()` statement of the color with `filament = …` gets the new value, otherwise the first one gets `, filament = n`; "automatic" removes the argument. Reason: the code stays the only source (CLI export and UI export agree, the agent sees the same), undo and the Git gutter capture the change; a preview override as in the parameter bar would be lost on saving and would be a second source of truth for the export.

4 Example model `examples/spec/mehrfarbig.scad`

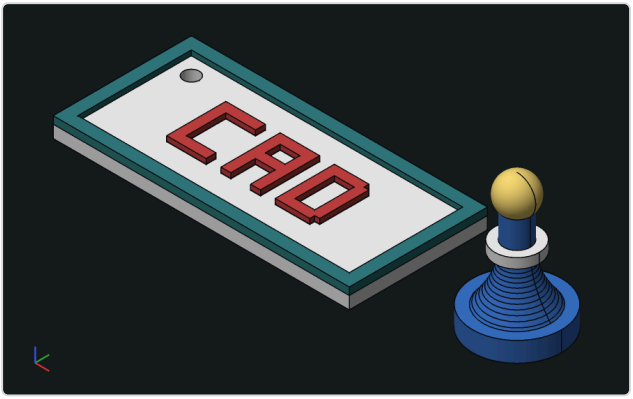
Door sign (white plate, frame teal, lettering "CAD" as relief in red with fixed slot 3, subtracted hanging hole) and game piece (blue body of base and revolved profile, white collar, yellow head) as two parts.



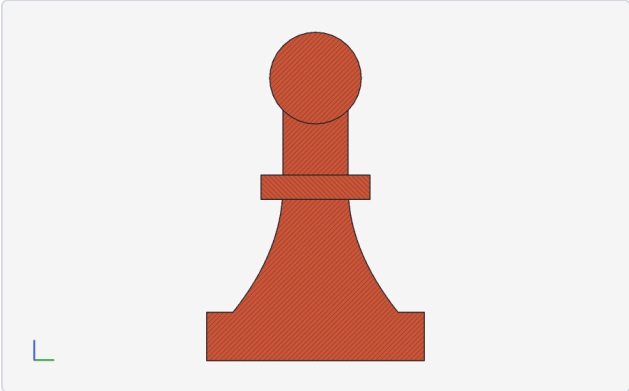
`cad render` – color per face



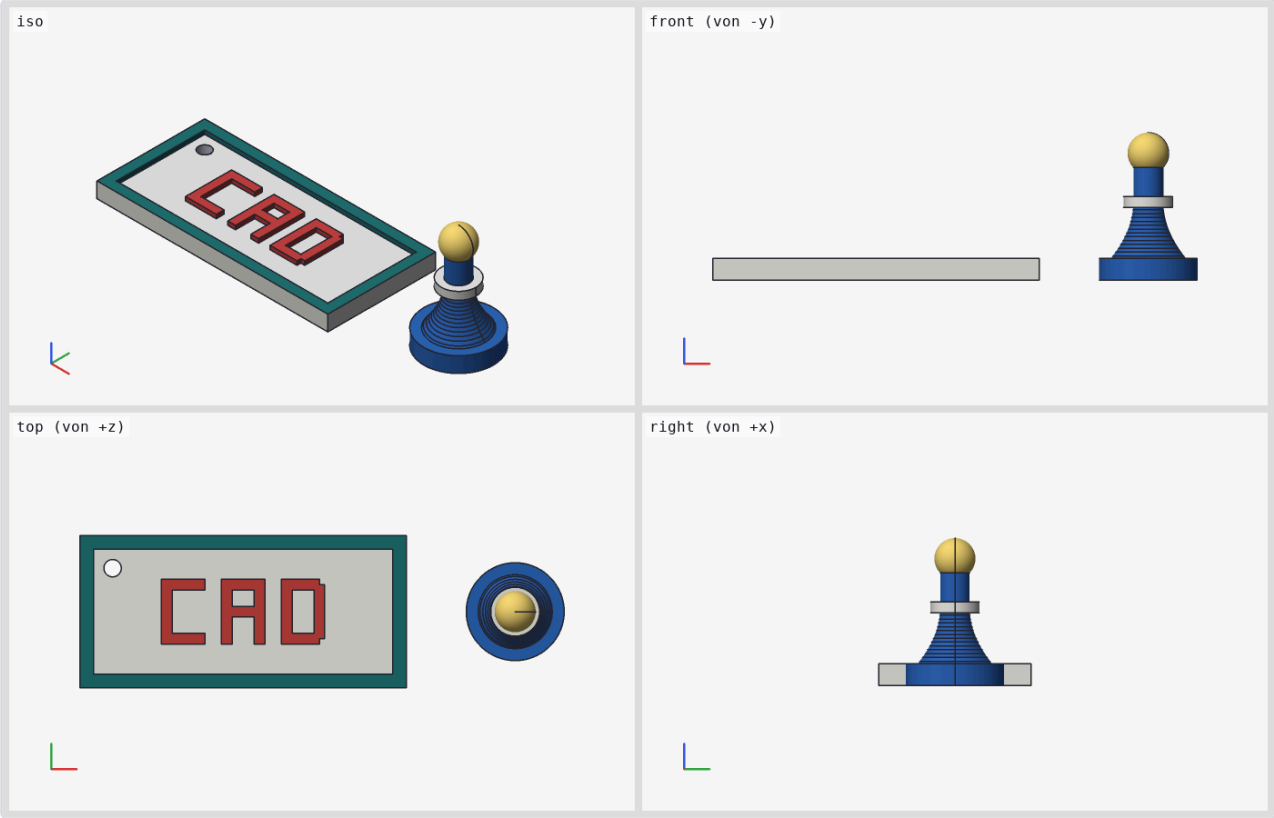
`cad render --theme dark`



`--show regions --theme dark`: bodies per color



`--section x=80 --part figur --show regions --view right`: section through the region bodies; the caps are hatched red as always, the contours show the boundaries between head (wins in the neck), collar and body



Contact sheet iso, front, top, right

Colors and volumes (`cad measure --colors`)

Slot	Color	Source	Lines	Volume	Mass PLA

F1	white `#f4f3f0`	palette	36, 66	4,900.8 mm³	6.08 g
F2	teal `#217274`	palette	37	622.5 mm³	0.77 g
F3 (filament = 3)	red `#c74240`	palette	42	208.8 mm³	0.26 g
F4	blue `#2e65b9`	palette	61	1,689.3 mm³	2.09 g
F5	yellow `#f7d974`	palette	67	226.2 mm³	0.28 g

Part	Regions (slot, color, mm³)	Sum	Part volume	Warning
schild	F1 white 4,773.57 · F2 teal 622.50 · F3 red 208.80	5,604.873	5,604.873	–
figur	F4 blue 1,689.33 · F1 white 127.23 · F5 yellow 226.24	2,042.797	2,042.797	–

▼ Source code

```
// Zweck: Mehrfarbig drucken (Bericht 11) – Türschild mit Schrift-Relief und Spielfigur als
// zwei Bauteile, Farben aus der Palette, ein fester Filament-Slot; Export als 3MF mit einem
// Teil je Farbe für Bambu Studio / OrcaSlicer.

// Blockbuchstaben aus Balken: s = Strichstärke, b × h = Feld des Buchstabens.
module balken(x, y, w, h, t) translate([x, y, 0]) cube([w, h, t]);

module buchstabe(c, b = 8, h = 12, s = 2, t = 1.2) {
    if (c == "C") {
        balken(0, 0, s, h, t);
        balken(0, h - s, b, s, t);
        balken(0, 0, b, s, t);
    } else if (c == "A") {
        balken(0, 0, s, h, t);
        balken(b - s, 0, s, h, t);
        balken(0, h - s, b, s, t);
        balken(0, h / 2 - s / 2, b, s, t);
    } else if (c == "D") {
        // Balken überlappen (nur Kanten berühren gäbe keinen geschlossenen Körper)
        balken(0, 0, s, h, t);
        balken(0, h - s, b - s / 2, s, t);
        balken(0, 0, b - s / 2, s, t);
        balken(b - s, s / 2, s, h - s, t);
    }
}

module schild(w = 60, d = 28, t = 3, rand = 2.5) {
    grid g {
        x: links = 0, schrift = 15, rechts = w;
        y: vorne = 0, zeile = 8, hinten = d;
        z: boden = 0, oben = t;
    }
    difference() {
        union() {
            // Grundplatte weiß, Rahmen petrol (sitzt auf der Platte)
            color("white") box(g[links, vorne, boden], g[rechts, hinten, oben]);
            color("teal") difference() {
                box(g[links, vorne, oben] - [0, 0, 0.5], g[rechts, hinten, oben] + [0, 0, 1]);
                box(g[links, vorne, oben] + [rand, rand, -1], g[rechts, hinten, oben] - [rand, rand, -2]);
            }
            // Schrift-Relief rot, auf Filament 3 festgelegt
            color("red", filament = 3)
                for (i = [0:2]) translate(g[schrift, zeile, oben] + [i * 11, 0, -0.2]) buchstabe(["C", "A",
"D"][i]);
        }
        // Aufhängebohrung: abgezogen, gilt für alle Farben
        translate(g[links, hinten, boden] + [6, -6, -1]) cylinder(h = t + 3, r = 1.6);
    }
}

module spielfigur(r = 9, h = 26) {
    grid p {
        x: achse = 0, hals = r * 0.3, rand = r * 0.85, fuss = r;
        y: boden = 0, sockel = 4, mitte = h * 0.55, kopf = h * 0.8;
    }
    // Profil: vom Rand des Sockels zum Hals (quadratisch), dann gerade bis zum Kopf
    z0 = p.sockel - 1;
```

```
    profil = concat(
        [[0, z0]],
        [for (i = [0:16]) let (t = i / 16) [p.hals + (p.rand - p.hals) * (1 - t) * (1 - t), z0 + t * (p.mitte -
z0)]],
        [[p.hals, p.kopf], [0, p.kopf]]);
    color("blue") {
        cylinder(h = p.sockel, r = p.fuss);
        rotate_extrude() polygon(profil);
    }
    // weißer Kragen und gelber Kopf: spätere Anweisungen gewinnen bei Überlappung
    color("white") translate([0, 0, p.mitte - 1]) cylinder(h = 2, r = r * 0.5);
    color("yellow") translate([0, 0, p.kopf + 2.5]) sphere(r * 0.42);
}

// Spielfigur neben dem Schild (eigenes Bauteil = eigenes Objekt im Slicer)
module figur(at = [80, 14, 0]) translate(at) spielfigur();

schild: schild();
figur: figur();
```

5 3MF for Bambu Studio and OrcaSlicer

Structure

```
Archive:  mehrfarbig.3mf
  Length      Date    Time    Name
---------  ---------- -----   ----
      371  1980-01-01 00:00   [Content_Types].xml
      260  1980-01-01 00:00   _rels/.rels
   603415  1980-01-01 00:00   3D/3dmodel.model
     1171  1980-01-01 00:00   Metadata/model_settings.config
---------                     -------
   605217                     4 files
```

- `3D/3dmodel.model` (core 2015/02 + materials extension): `m:colorgroup` with one `m:color` per color (#RRGGBBAA); per (part, color) one mesh object with `pid`/`pindex` (color at object level); per part one object with `components` without `transform` (all meshes in the same coordinate system – nothing is moved); one build item per part. `Application` = "controller-cad 0.1.0".
- `Metadata/model_settings.config`: per part `<object id>` (= build item) with `name` and `extruder` (slot of the largest color), per color `<part id>` (= ID of the mesh object) with `name` and `extruder` = slot.
- No `Metadata/project_settings.config` (see assumptions). Content type for `.config` registered.

```
<?xml version="1.0" encoding="UTF-8"?>
<model unit="millimeter" xml:lang="en-US" xmlns="http://schemas.microsoft.com/3dmanufacturing/core/2015/02"
xmlns:m="http://schemas.microsoft.com/3dmanufacturing/material/2015/02">
 <metadata name="Application">controller-cad 0.1.0</metadata>
 <resources>
  <m:colorgroup id="1">
   <m:color color="#F4F3F0FF"/>
   <m:color color="#217274FF"/>
   <m:color color="#C74240FF"/>
   <m:color color="#2E65B9FF"/>
   <m:color color="#F7D974FF"/>
  </m:colorgroup>
     <vertex x="6.8755355" y="23.339193" z="0"/>
     <vertex x="60" y="28" z="0"/>
     …
  <object id="5" type="model" name="schild">
   <components>
    <component objectid="2"/>
    <component objectid="3"/>
    <component objectid="4"/>
   </components>
  …
 <build>
  <item objectid="5"/>
  <item objectid="9"/>
 </build>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<config>
  <object id="5">
    <metadata key="name" value="schild"/>
    <metadata key="extruder" value="1"/>
    <part id="2" subtype="normal_part">
      <metadata key="name" value="schild – white"/>
      <metadata key="extruder" value="1"/>
    </part>
    <part id="3" subtype="normal_part">
      <metadata key="name" value="schild – teal"/>
      <metadata key="extruder" value="2"/>
    </part>
    <part id="4" subtype="normal_part">
      <metadata key="name" value="schild – red"/>
      <metadata key="extruder" value="3"/>
    </part>
  </object>
  <object id="9">
    <metadata key="name" value="figur"/>
    <metadata key="extruder" value="4"/>
    <part id="6" subtype="normal_part">
      <metadata key="name" value="figur – blue"/>
      <metadata key="extruder" value="4"/>
    </part>
    <part id="7" subtype="normal_part">
      <metadata key="name" value="figur – white"/>
      <metadata key="extruder" value="1"/>
    </part>
    <part id="8" subtype="normal_part">
      <metadata key="name" value="figur – yellow"/>
      <metadata key="extruder" value="5"/>
    </part>
  </object>
</config>
```

Assumptions and their sources

Read in the source code of Bambu Studio (`bambulab/BambuStudio`, branch master) and OrcaSlicer (`SoftFever/OrcaSlicer`, branch main), retrieved on 2026-10-23 via raw.githubusercontent.com (checksums below):

1. **Foreign 3MF:** Bambu Studio treats a file as its own only if `Application` starts with "BambuStudio-" (`_handle_end_metadata`, OrcaSlicer additionally "OrcaSlicer-"). We do not pose as Bambu Studio; the file is loaded as "From_Other" (`Model::read_from_archive`; PrusaSlicer files are recognized by `PrusaFileParser` from the name "PrusaSlicer").
2. **`model_settings.config` is also read for foreign files** (`_load_model_from_file`, branch `BBS_MODEL_CONFIG_FILE`; errors are fatal only for its own files). `<object id>` is matched via the ID of the build item object, `<part id>` via the ID of the sub-object (`_handle_start_config_volume`: without a triangle range, `subobject_id` counts; `_generate_volumes_new` looks for `volume_iter.subobject_id == sub_object->id`). All other keys of a `part` go into the volume configuration (`volume->config.set_deserialize`), including `extruder`.
3. **Components without `p:path`** refer to objects of the same file (`_handle_start_component`: empty path = current model); `_generate_current_object_list` resolves them into sub-objects, each component becomes a sub-part (volume) of the same object with the component's transformation (none here).
4. **One volume:** after loading, the importer removes the `extruder` of a single volume; then the object's applies – that is why we also write it on the object.
5. **Colors:** the parser compares element names literally with `m:colorgroup`/`m:color` (prefix `m` required). With `pid`/`pindex` on the mesh object every triangle gets the color (`_handle_start_triangle`), and current Bambu Studio offers the color dialog on opening (`Plater.cpp`: `extract_colors_to_obj_dialog`, `ObjColorDialog`); "Append" adds the colors as filaments, and with colors per volume `ObjColorPanel::apply_volume_filament_ids` sets the extruders per volume without painting over the volume boundaries. Older versions and OrcaSlicer use the color group only at object level and without metadata.
6. **No `project_settings.config`:** Bambu Studio loads it only from its own files (`dont_load_config = true` without generator version); OrcaSlicer always loads it and applies it over its defaults (`config += config_loaded`) – a file with only `filament_colour` would reset printer and process settings there. The filament colors therefore come from the color dialog (Bambu) or from the user/AMS sync.

7. **Missing filaments:** on import (geometry only) Bambu Studio extends the filament list up to the highest `extruder` in the file (`Plater.cpp`: "add extruder ... if only import geometry", with colors from `get_next_color_for_filament`).

```
b725d2c7d872c7e4ba31fa6d67c333d59da939e417d02eae16d8e9a15fa16608  SoftFever/OrcaSlicer
main/src/libslic3r/Format/bbs/3mf.cpp
c32c66d03d16fe17cf5f4cde3ac68025791b20f423984aac4694094d6033d650  SoftFever/OrcaSlicer
main/src/slic3r/GUI/Plater.cpp
21a31bca20d5de47ac350342fcaaa4794ed5d1b2157869b2d6afe3f145a5c4dd  bambulab/BambuStudio
master/src/libslic3r/Format/3mf.cpp
b5da58b9404d8bdb6b14f0920e63fe16e2a94175f78056cece44074011e0056c  bambulab/BambuStudio
master/src/libslic3r/Format/bbs/3mf.cpp
84843d5f4910909abc37403ad2dfc1ff404a51c134f8a207d55ab571d1223792  bambulab/BambuStudio
master/src/libslic3r/Model.cpp
c6f9aa6c45c04c2849118a11bc0c0c81329aec9cf9817eb01edd7733210f2765  bambulab/BambuStudio
master/src/slic3r/GUI/ObjColorDialog.cpp
4a5a4db32d9ae48e98ba0e53c1fba5cf6700eb67cb2746318627b43c4add86d8  bambulab/BambuStudio
master/src/slic3r/GUI/Plater.cpp
```

6 Verification of the 3MF

`tools/report/verify3mf.py` on the file from `cad export examples/spec/mehrfarbig.scad --part schild --part figur -o mehrfarbig.3mf`, expected values from `cad measure --colors --json`:

```
lib3mf 2.5.0: 6 Netze, 2 Komponentenobjekte, 2 Build-Items, 5 Farben, Warnungen 0, alle mannigfaltig: True
trimesh 5.1.1: 6 Geometrien, wasserdicht: True
Bambu/Orca-Import (Modell): Objekt schild: schild — white → Extruder 1 #F4F3F0FF, schild — teal → Extruder 2
#217274FF, schild — red → Extruder 3 #C74240FF
Bambu/Orca-Import (Modell): Objekt figur: figur — blue → Extruder 4 #2E65B9FF, figur — white → Extruder 1
#F4F3F0FF, figur — yellow → Extruder 5 #F7D974FF
PrusaSlicer (PrusaSlicer-2.7.2+UNKNOWN based on Slic3r (with GUI support)): 6 Objekte, Exit 0
  schild white: F1 Bambu 1, B-Rep 4773.57 Netz 4773.64 mm³ ok
  schild teal: F2 Bambu 2, B-Rep 622.50 Netz 622.50 mm³ ok
  schild red: F3 Bambu 3, B-Rep 208.80 Netz 208.80 mm³ ok
  figur blue: F4 Bambu 4, B-Rep 1689.33 Netz 1688.17 mm³ ok
  figur white: F1 Bambu 1, B-Rep 127.23 Netz 127.03 mm³ ok
  figur yellow: F5 Bambu 5, B-Rep 226.24 Netz 225.32 mm³ ok
alle Prüfungen bestanden
```

Part	Color	Slot	Extruder (Bambu model)	Color in 3MF	Volume B-rep	Volume mesh	
schild	white	F1	1	#F4F3F0FF	4,773.573	4,773.635	ok
schild	teal	F2	2	#217274FF	622.500	622.500	ok
schild	red	F3	3	#C74240FF	208.800	208.800	ok
figur	blue	F4	4	#2E65B9FF	1,689.325	1,688.168	ok
figur	white	F1	1	#F4F3F0FF	127.235	127.029	ok
figur	yellow	F5	5	#F7D974FF	226.237	225.322	ok

- **lib3mf** 2.5.0 (library of the 3MF Consortium): reads without warning, 6 meshes, all manifold and oriented (`IsManifoldAndOriented`), components and build items without transformation, color per mesh from the color group.
- **trimesh** 5.1.1: 6 geometries, all watertight.
- **Bambu/Orca model**: the importer re-implemented (section 5): build item → components → sub-objects, sub-parts via `part id`, extruder per sub-part, color per sub-part; file not recognized as a Bambu project (intended), no `project_settings.config`. Volume B-rep ↔ mesh within 0.5 %.
- **PrusaSlicer** PrusaSlicer-2.7.2+UNKNOWN based on Slic3r (with GUI support) (apt, CLI `--info`): loads the file, 6 objects (one per mesh, at the right position; PrusaSlicer does not read `model_settings.config` and turns components of foreign files into separate objects), all manifold, volumes as above.
- **Bambu Studio / OrcaSlicer themselves** were not run: the container has no access to their release files (AppImage), building from source would take hours and many GB. The mapping is therefore derived from the source code and checked with the model; section 10 describes the check in the slicer.
- STL per color (ZIP, `cad export --per-color`):

```
Archive:  mehrfarbig-farben.zip
  Length      Date    Time    Name
```

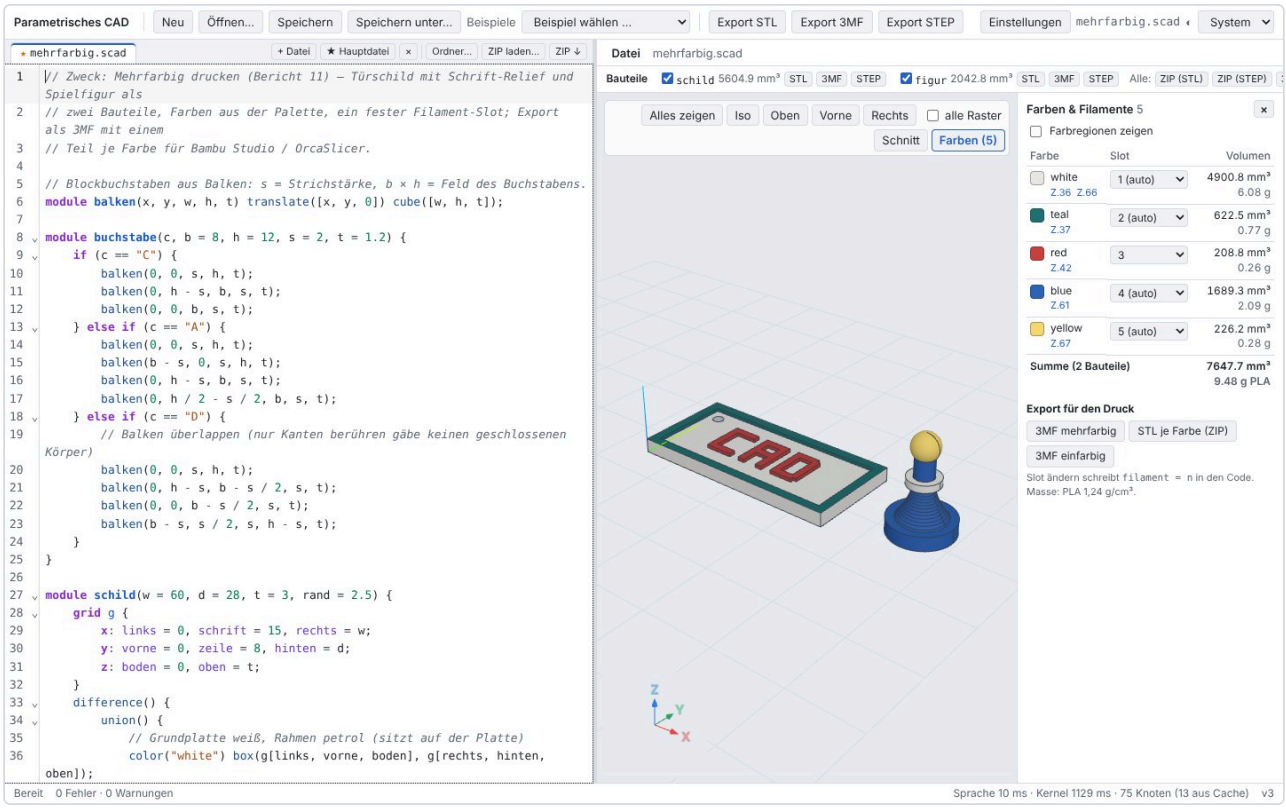
```
---------  ---------- -----   ----
    19284  1980-01-01 00:00   schild_F1_white.stl
     1684  1980-01-01 00:00   schild_F2_teal.stl
     6284  1980-01-01 00:00   schild_F3_red.stl
   196584  1980-01-01 00:00   figur_F4_blue.stl
    12684  1980-01-01 00:00   figur_F1_white.stl
   185684  1980-01-01 00:00   figur_F5_yellow.stl
---------                     -------
   422204                     6 files
```

7 UI: colors & filaments (Playwright, Chromium)

F01 Color list with slots, volume, mass

Example mehrfarbig.scad: five colors in the order of first occurrence, red fixed with filament = 3, the others automatic (1, 2, 4, 5). Volume per color from the color regions of both parts, mass for PLA 1.24 g/cm³. In the 3D view each face carries the color of its color() call.

- ✓ labels and slots white 1, teal 2, red 3 (explicit), blue 4, yellow 5
- ✓ sum of the regions = volume per part
- ✓ red 208.8 mm³ / 0.26 g; swatches with palette values
- ✓ 1072 red and 6499 teal pixels in the view



light theme, color list on the right

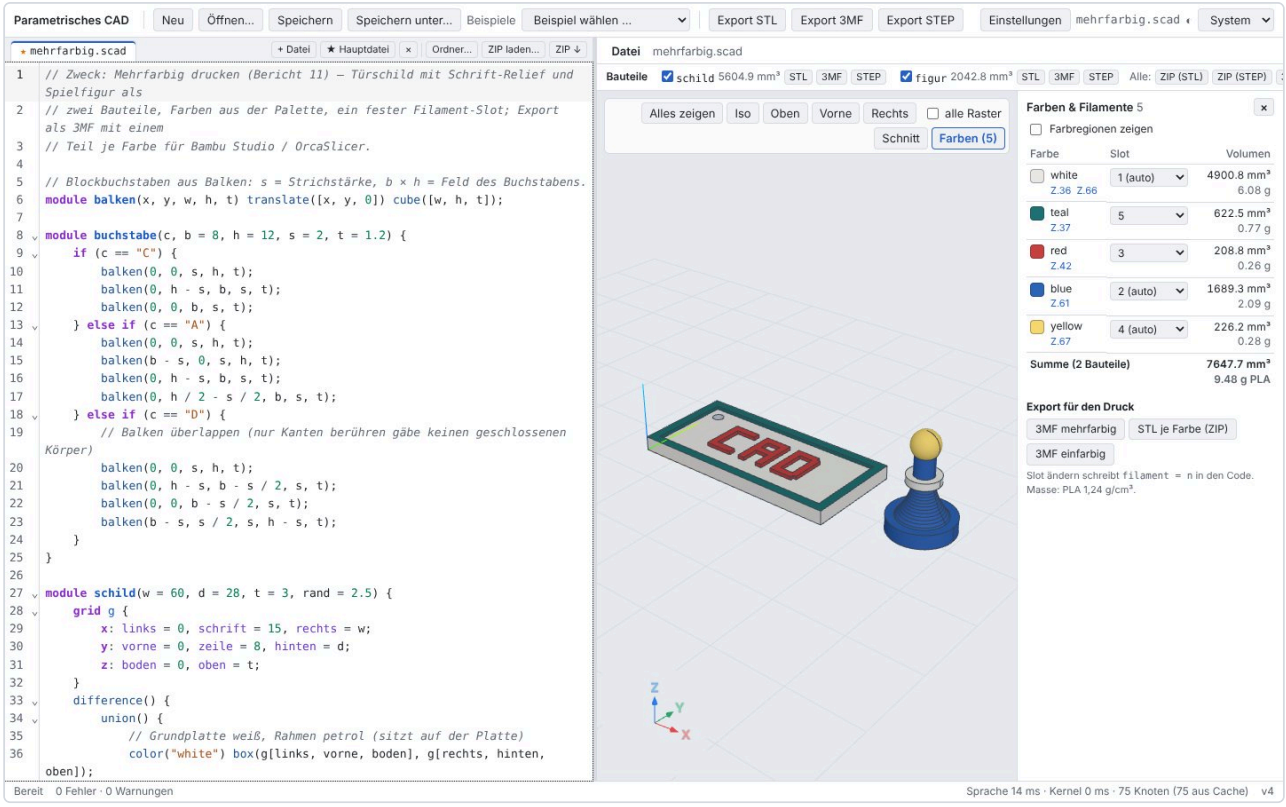


Color list

F02 Slot via dropdown

Slot of teal set to 5: the editor gets color("teal", filament = 5) (text edit via the engine, undoable), blue moves to the free slot 2. "automatic" (German UI: „automatisch“) removes the argument again. Material without color() is always filament 1.

- ✓ text contains color("teal", filament = 5)
- ✓ slots afterwards white 1, teal 5, red 3, blue 2, yellow 4
- ✓ automatic: text as before again
- ✓ default entry without selector



teal on slot 5, code changed

F03 Export for Bambu Studio / OrcaSlicer

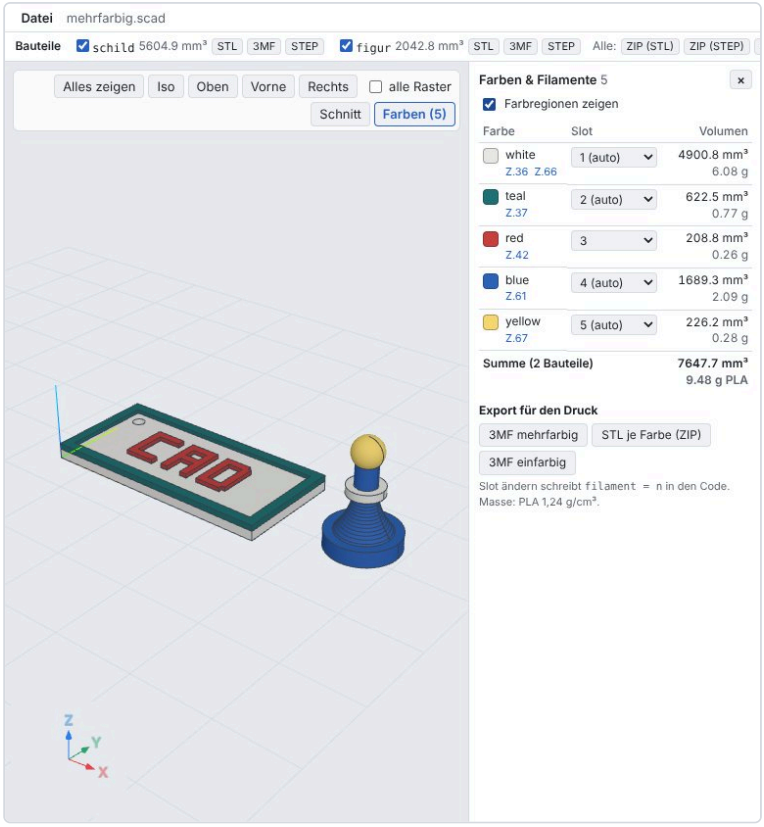
"3MF multi-color" (German UI: „3MF mehrfarbig“): one object per part (schild, figur), in it one sub-part per color without offset, color group with five colors, Metadata/model_settings.config with extruder per sub-part. "STL per color" („STL je Farbe“): ZIP with six files F<slot>_<farbe>.stl. "3MF single-color" („3MF einfarbig“) without colors.

- ✓ 2 build items, 6 components, 5 m:color
- ✓ schild – red → extruder 3
- ✓ ZIP with 6 STL (Bauteil_F<slot>_<farbe>.stl)
- ✓ single-color without colorgroup

F04 Color regions in the view

"Show color regions": instead of the color per face, the bodies per color as they are exported (six bodies: three per part).

- ✓ 6 region bodies drawn
- ✓ 532 yellow pixels
- ✓ off: color per face again

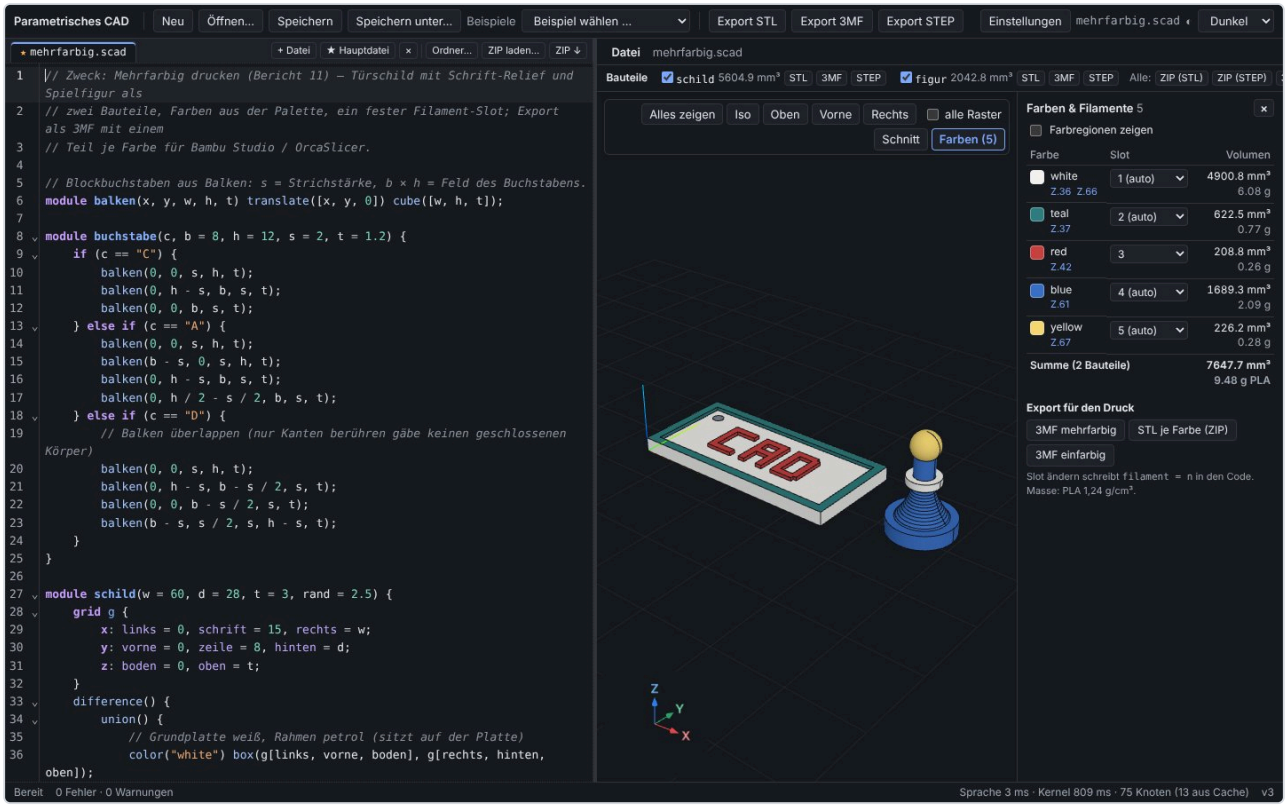


Color regions

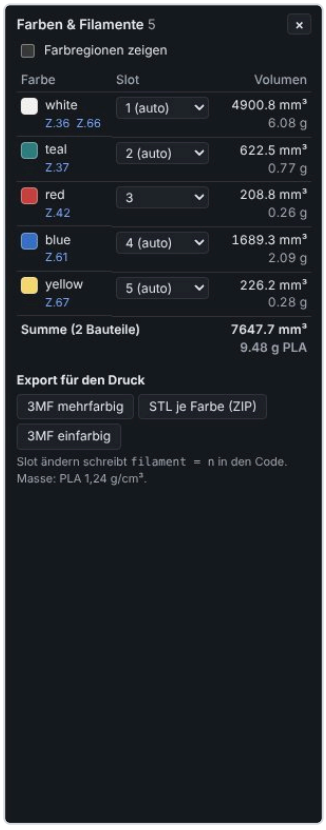
F05 Dark theme

Theme "Dark" (German UI: „Dunkel“): panels, editor (syntax, line numbers, search bar), status bar and color list dark, text contrast ≥ 4.5 (computed styles); 3D view with dark background, palette colors in the dark variant (teal #307e80 instead of #217274), export uses the base values.

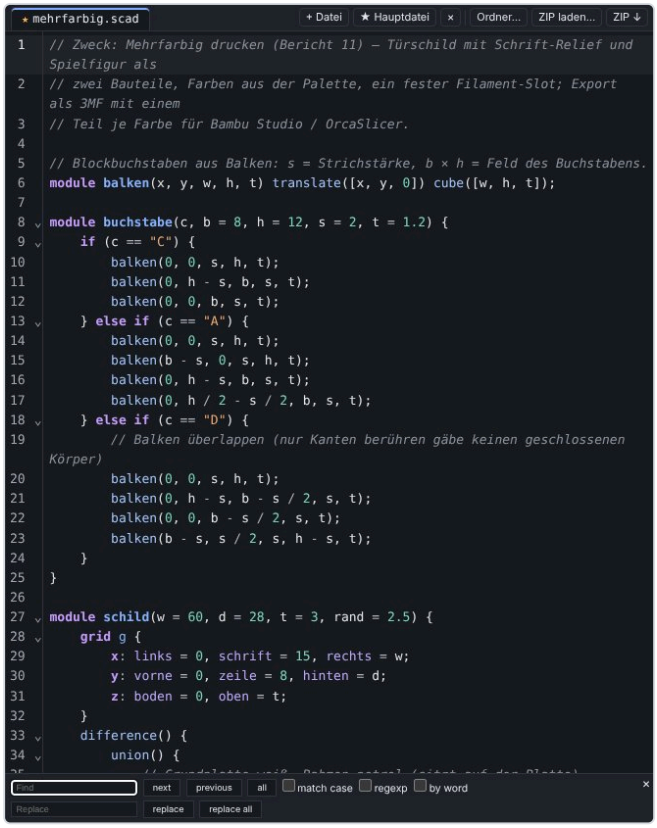
- ✓ text on background: 15.15
- ✓ editor text: 14.19
- ✓ line numbers: 7.37
- ✓ status bar: 7.37
- ✓ color list: 14.19
- ✓ toolbar: 12.80



dark theme



Color list dark



Editor with search bar

F06 Theme "System"

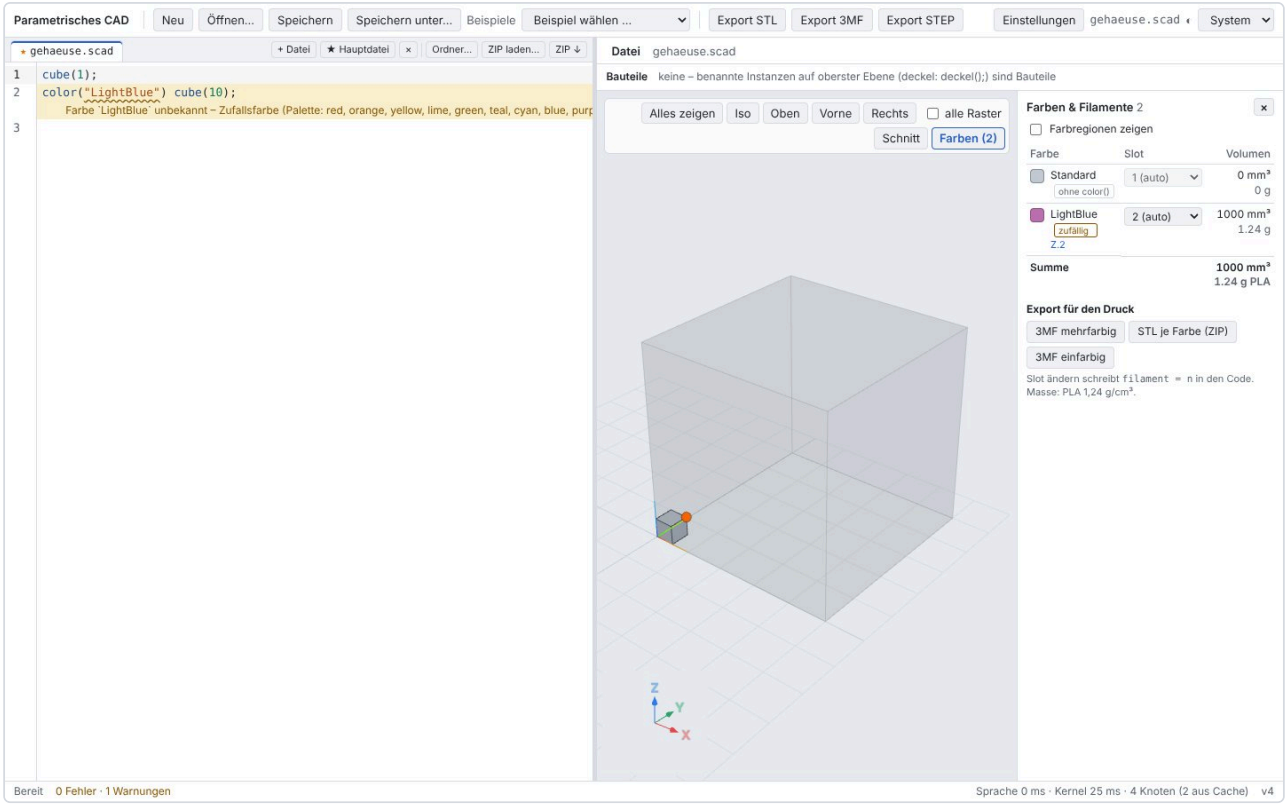
The default is "System": the theme follows prefers-color-scheme, also when switching at run time; "Light"/"Dark" (German UI: „Hell“/„Dunkel“) are fixed and persist after reloading.

- ✓ System + dark → dark, then light → light, dark → dark
- ✓ fixed Light stays with dark system
- ✓ choice survives reload

F07 Unknown color

color("lightblue"): warning on the argument „Farbe `lightblue` unbekannt – Zufallsfarbe (Palette: …)“, the color comes from the hash of the name (the same for LightBlue, same lightness and saturation as the palette); "Teal" (case-insensitive) is the palette color.

- ✓ random color #bd70af for lightblue and LightBlue
- ✓ warning in line 1
- ✓ Teal = palette

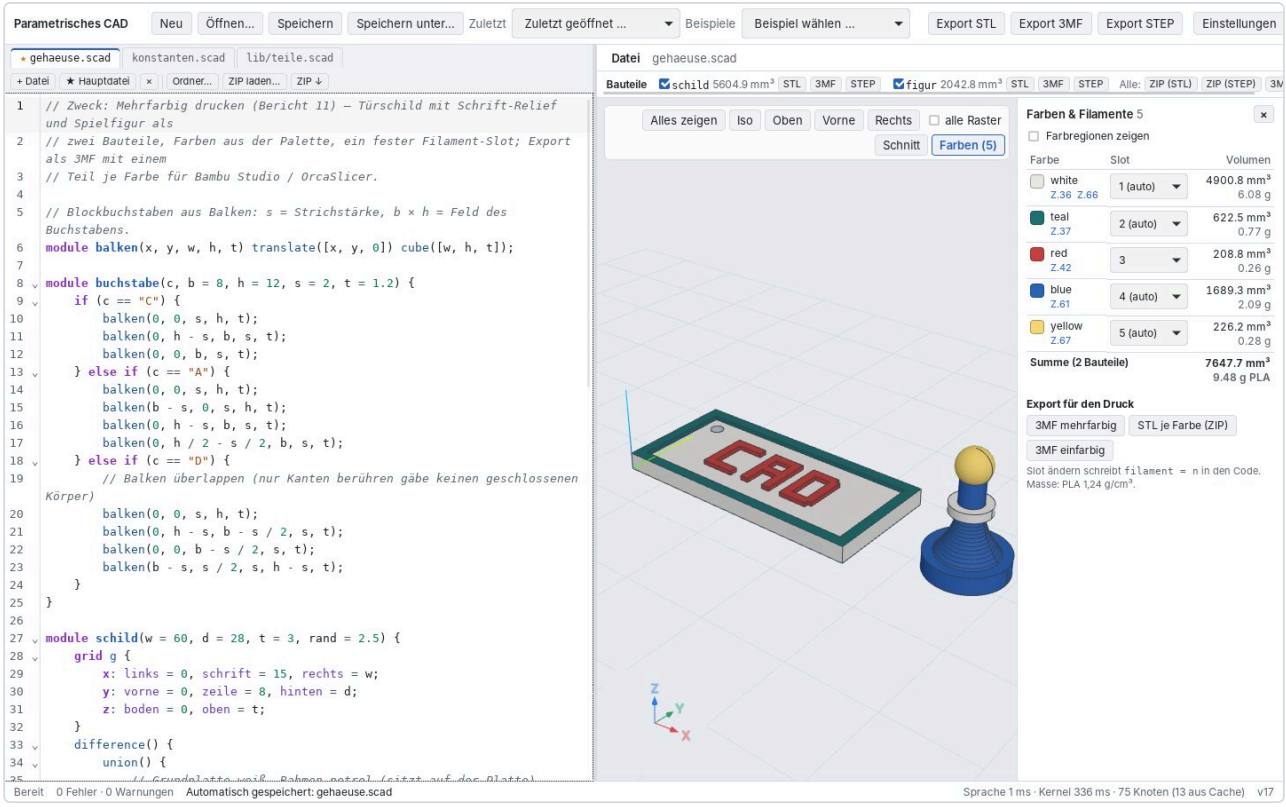


Warning and color list

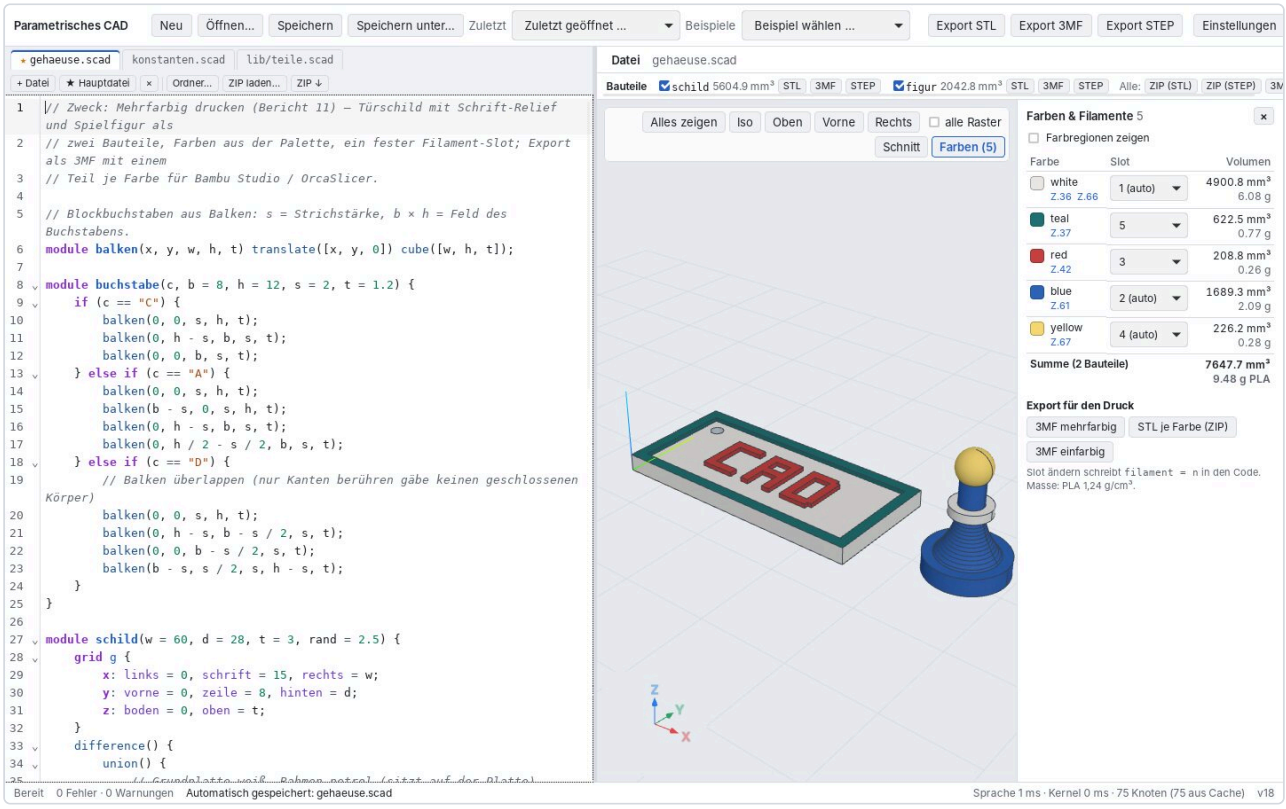
Desktop app (Tauri, native kernel)

T12 Colors & filaments in Tauri: list, change slot, 3MF multi-color (native kernel)

- ✓ colors white:1, teal:2, red:3, blue:4, yellow:5
- ✓ schild: sum 5604.873 = 5604.873 mm³
- ✓ figur: sum 2042.797 = 2042.797 mm³
- ✓ teal fixed on slot 5
- ✓ file marked as modified
- ✓ 3MF: {"files":
  [],"mColors":5,"colorGroup":true,"modelSettings":true,"items":2,"colors":true,"components":6,"bytes":605695,"extrud
  [1,1,5,3,2,2,1,4],"triangles":8434}
- ✓ extruder per sub-part 1,1,5,3,2,2,1,4
- ✓ STL per color: schild_F1_white.stl, schild_F5_teal.stl, schild_F3_red.stl, figur_F2_blue.stl, figur_F1_white.stl, figur_F4_yellow.stl



Color list in the desktop app (light theme)

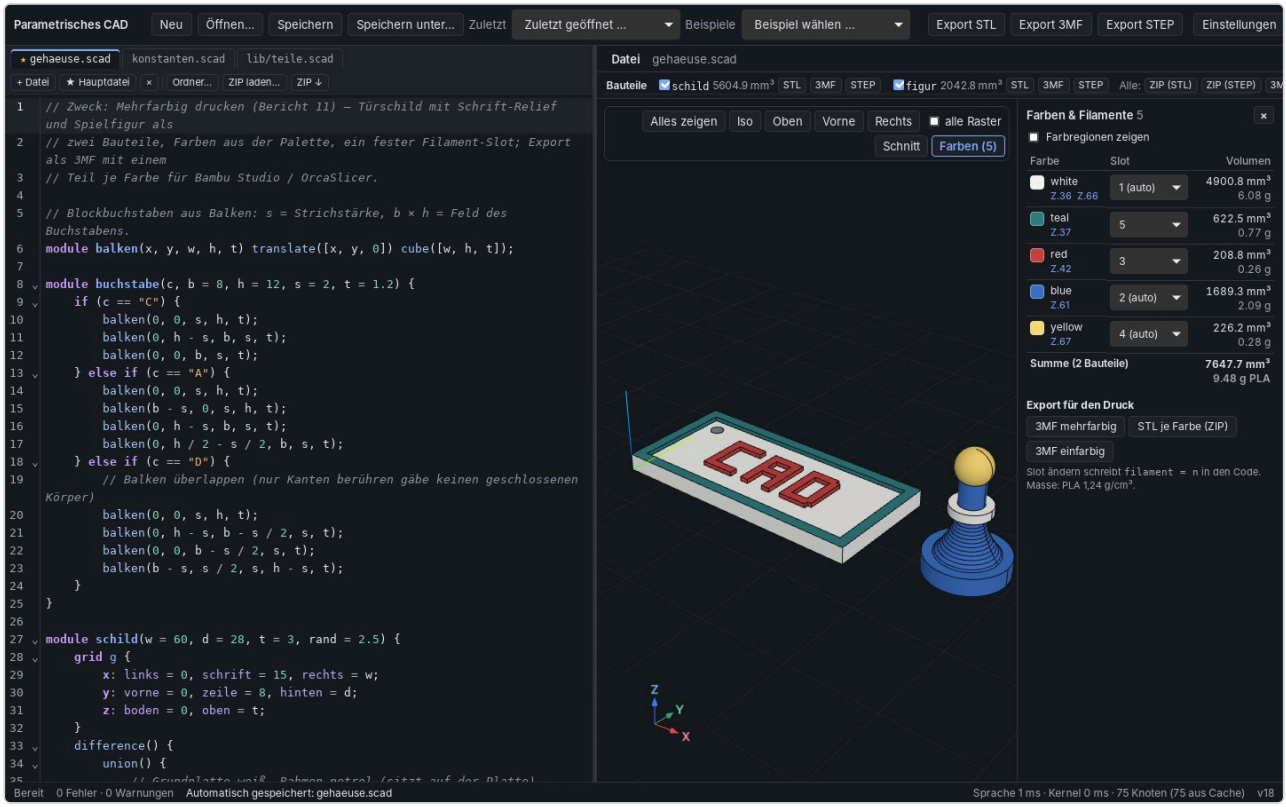


teal on slot 5: code changed

T13 Dark theme in Tauri: contrast (computed styles), view, editor

- ✓ text: contrast 15.15 ≥ 4.5
- ✓ editor: contrast 14.19 ≥ 4.5
- ✓ status bar: contrast 7.37 ≥ 4.5
- ✓ color list: contrast 14.19 ≥ 4.5
- ✓ toolbar: contrast 12.80 ≥ 4.5
- ✓ teal dark variant (rgb(48, 126, 128))

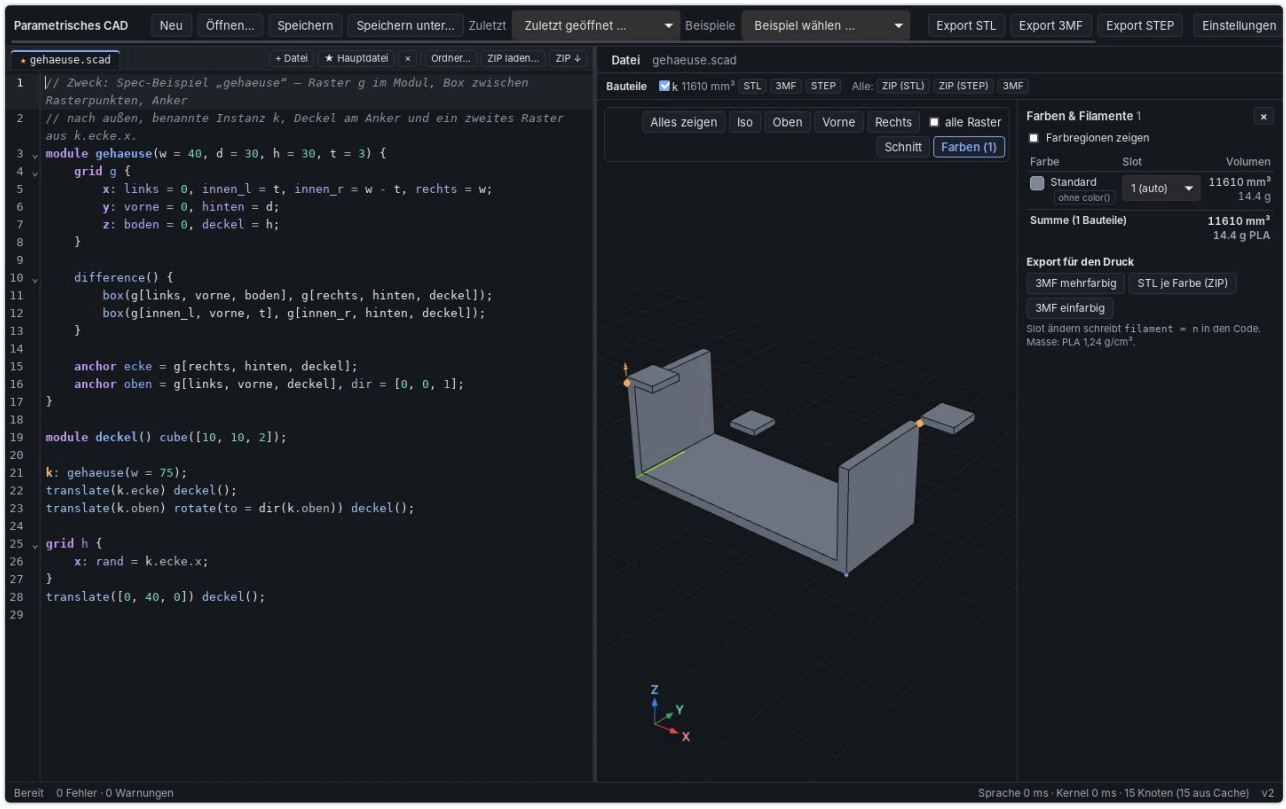
✓ color per face in the view



dark theme in the desktop app

T14 Theme "System" in Tauri follows the GTK color scheme (prefers-color-scheme)

✓ System (light) → data-theme light
✓ WebKitGTK reports prefers-color-scheme: dark with GTK_THEME=Adwaita:dark (true)
✓ theme System → dark



Theme System with dark GTK scheme

8 Dark theme

- **Choice** in the toolbar: System (default; follows `prefers-color-scheme` with a listener, also when switching at run time), Light, Dark; stored in `localStorage` (`cad-theme`). In Tauri, WebKitGTK reports the GTK color scheme (test T14 with `GTK_THEME=Adwaita:dark`).
- **Tokens** per theme in `styles.css`: surfaces, text, accent, error/warning (also line backgrounds), syntax, selection, hover, diff view, shadows, dialog backdrop, swatches; 3D view (background, floor and grid, model, edges, highlight, dimming, zones, subtracted bodies, section cap, handles including fill, anchors, snap with its own text color), Git gutter. The 3D view re-reads the tokens on a switch.
- **Editor components with their own default theme** (CodeMirror: autocompletion, search bar, matches, bracket pairs, fold markers) now follow the tokens; before, they stayed light in the dark theme.
- **Check:** `web/src/logic/contrast.test.ts` computes the WCAG contrast from the tokens for all pairs (translucent backgrounds blended over the surface): text ≥ 4.5:1, UI components ≥ 3:1; F05 and T13 measure the same pairs via `getComputedStyle` in Chromium and WebKitGTK, F05 additionally pixels of the 3D view (background, palette variant).

Pair	Token	light	dark	Target
Text	`--text` on `--panel`	15.80	14.19	4.5
Muted text	`--muted` on `--panel`	5.80	7.37	4.5
Muted text on surface 2	`--muted` on `--panel-2`	5.17	6.65	4.5
Accent (links, lines of the color list)	`--accent` on `--panel`	5.71	7.47	4.5
Error	`--error` on `--panel`	5.52	6.89	4.5
Error on error line	`--error` on `--error-bg`	4.66	5.62	4.5
Warning on warning line	`--warning` on `--warning-bg`	5.48	7.20	4.5
Syntax: keyword	`--hl-keyword` on `--panel`	6.27	8.19	4.5
Syntax: number	`--hl-number` on `--panel`	5.13	10.27	4.5
Syntax: string	`--hl-string` on `--panel`	5.65	9.16	4.5
Syntax: comment	`--hl-comment` on `--panel`	5.40	5.76	4.5
Syntax: comment in active line	`--hl-comment` on `--active-line`	5.05	5.11	4.5
Syntax: definition	`--hl-def` on `--panel`	6.27	8.27	4.5
Syntax: modifier	`--hl-modifier` on `--panel`	5.68	7.57	4.5
Text on selection	`--text` on `--accent-soft`	13.34	10.17	4.5
Text on diff (new)	`--text` on `--diff-add`	13.32	10.30	4.5
Text on diff (deleted)	`--text` on `--diff-del`	13.17	11.31	4.5
Text on accent	`--on-accent` on `--accent`	5.71	8.10	4.5
Snap label	`--on-snap` on `--view-snap`	5.38	9.00	4.5
Git gutter new (UI, ≥ 3)	`--git-added` on `--panel`	3.49	7.36	3.0
3D: edges on model (≥ 3)	`--view-edge` on `--view-model`	6.34	5.38	3.0
3D: grid on background (≥ 3)	`--view-grid` on `--view-bg`	5.35	6.78	3.0

9 Tests

- `cargo test --workspace`: 388 passed, 0 failed, 0 ignored; `cargo clippy --workspace --all-targets -- -D warnings`: no findings; `cargo fmt --all --check`: no findings.
- Web: `pnpm typecheck`, Vitest 193 (new: `colors.test.ts`, `contrast.test.ts`), Playwright 84/84 (all existing scenarios and F01–F07).
- Tauri E2E 14/14 (T01–T11 as in report 9, new T12–T14).
- Found and fixed during the runs: (1) the color list is open by default and requested regions for every finished scene; for parts without color this cost a volume computation in the kernel, which with the watchdog set to 1.5 s

in C19 triggered a second restart – now without a kernel call (cached part volume). (2) T08: with the color list open, the view bar wraps and the section bar lay over the second row ("element click intercepted") – both are now stacked in one container. (3) Intermittent failure in `supervisor_watchdog_replaces_engine_and_blocks_key` (1 in 5 runs, already before this task): the input thread of the desktop engine passed on the messages of a frame one by one, so `SetSource` and `__debugHang` could fall into two iterations of the engine loop and a different node was blocked; a frame is now one channel message (8/8 runs green). (4) P07 expected the old single-color 3MF of the part bar (now multi-color: build items counted instead of objects).

- OpenSCAD comparison `compare.py --eval`: 33/33 (none of the 33 models uses `color()`; geometry unchanged); STEP `verify.py` with OpenCASCADE: 48/48 passed.

Test	Checks	Result
`display_variants` cad-eval/tests/palette.rs	display variants: same hue, lightness only where needed, contrast to the view background	ok
`children_inherit_the_colour` cad-eval/tests/palette.rs	children inherit the color, inner `color()` wins	ok
`filament_parameter` cad-eval/tests/palette.rs	`filament = n`: range 1–16, integer, contradiction warns, not positional	ok
`hex_vectors_and_alpha_stay_exact` cad-eval/tests/palette.rs	`#rgb`, `#rrggbbaa`, vectors, `alpha` exact; broken hex warns	ok
`palette_ts_is_current` cad-eval/tests/palette.rs	generated table `web/src/logic/palette.ts` up to date	ok
`palette_names_case_insensitive` cad-eval/tests/palette.rs	names case-insensitive, `grey` = gray, palette values	ok
`unknown_name_warns_and_gets_a_stable_random_colour` cad-eval/tests/palette.rs	warning on the argument, stable random color (hash), L/C as palette, different hues, no German aliases	ok
`palette_is_distinguishable` cad-eval/tests/palette.rs	ΔE_OK ≥ 0.10 normal, ≥ 0,065 deuteranopia, ≥ 0.05 protanopia, ≥ 0,025 tritanopia; chroma ≤ 0.18, no pure colors	ok
`unchanged_value_gives_no_edit` cad-eval/tests/filament_edit.rs	same value: no edit	ok
`slot_is_added_to_the_first_statement` cad-eval/tests/filament_edit.rs	without slot: `, filament = n` on the first statement of the color	ok
`existing_slots_are_replaced_or_removed` cad-eval/tests/filament_edit.rs	existing slots replaced (all spellings of the color), "automatic" removes them	ok
`filament_slots` cad-engine/tests/colors.rs	explicit slots, lowest free slot, same color in three spellings = one entry	ok
`later_statement_wins_overlaps` cad-engine/tests/colors.rs	overlap: later statement wins (500 / 1000 mm³)	ok
`multicolour_3mf_structure` cad-engine/tests/colors.rs	3MF: files, color group, objects, components without transformation, `model_settings.config` (part ID = mesh ID, extruder), STL per color	ok
`nested_colour_wins_over_its_parent` cad-engine/tests/colors.rs	inner `color()` wins over the outer one	ok
`region_meshes_share_coordinates` cad-engine/tests/colors.rs	meshes of the regions: volume, shared coordinates	ok
`resize_scales_the_colour_material` cad-engine/tests/colors.rs	`resize` when placing the color bodies	ok
`fillet_between_colours_goes_to_the_first_coloured_subtree` cad-engine/tests/colors.rs	fillet between two colors → first color; volume sum	ok
`regions_are_cached_and_colour_changes_recompute_no_nodes` cad-engine/tests/colors.rs	regions cached (0 kernel calls), color/slot change recomputes no node	ok
`transforms_and_modules` cad-engine/tests/colors.rs	modules, loops, rotations	ok

`uncoloured_material_is_the_default_colour_on_filament_1` cad-engine/tests/colors.rs	uncolored material = default, slot 1	ok
`subtracted_material_counts_for_all_colours` cad-engine/tests/colors.rs	subtracted material applies to all colors; color only in the subtraction: no slot	ok
`uncoloured_part_needs_no_boolean` cad-engine/tests/colors.rs	without color nodes: one default region without boolean, volume = part volume, mesh with all bodies	ok
`colour_regions_with_meshes` cad-session/tests/colors.rs	`RequestColorRegions` with meshes, old version is discarded	ok
`scene_lists_colours_with_slots` cad-session/tests/colors.rs	protocol: `Scene.colors` (also preliminary scene), warning at line 3	ok
`multicolour_exports` cad-session/tests/colors.rs	export 3MF (multi-color by default), `colors: false`, STL per color	ok
`set_filament_writes_the_code` cad-session/tests/colors.rs	`SetFilament` → edits, undo, error cases	ok
`measure_colours` cad-cli/tests/colors.rs	`cad measure --colors`: volume, mass 1.24 g/cm³, parts	ok
`outline_lists_colours_slots_and_volumes` cad-cli/tests/colors.rs	`cad outline`: colors, slots, volumes, sum per part, without kernel no volumes	ok
`render_theme_and_regions` cad-cli/tests/colors.rs	`cad render --theme dark --show regions`: dark background (pixels), 6 region bodies	ok
`export_multicolour_single_and_per_colour` cad-cli/tests/colors.rs	`cad export` multi-color, `--single-color`, `--per-color`, STL single-color	ok

Tauri E2E	Scenario	Duration	Result
T01	App starts with a file, scene appears (native kernel)	2.2 s	ok
T02	External change without unsaved changes: silently reloaded, undo as a separate step	0.8 s	ok
T03	Git markers in the gutter (new, changed, deleted)	1.0 s	ok
T04	External change with unsaved changes: notification bar, Compare, Keep mine, save without loop	2.5 s	ok
T05	Deleted and renamed file	1.3 s	ok
T06	Agent workflow: change file from the shell, `cad check`, UI follows (image sequence)	1.1 s	ok
T07	`cad ui status` / `show` / `screenshot` against the running app	1.0 s	ok
T08	Section view in Tauri (cap, contour, dragging, CLI call)	3.9 s	ok
T09	Watchdog in Tauri: hanging kernel → engine process replaced, node blocked	8.0 s	ok
T10	Folder as project, auto-save	3.3 s	ok
T11	Keystroke → scene, native (measurement)	11.3 s	ok
T12	Colors & filaments in Tauri: list, change slot, 3MF multi-color (native kernel)	3.4 s	ok
T13	Dark theme in Tauri: contrast (computed styles), view, editor	0.6 s	ok
T14	Theme "System" in Tauri follows the GTK color scheme (prefers-color-scheme)	4.2 s	ok

10 Instructions: open in Bambu Studio

> There was no Bambu Studio in the container; please verify these steps yourself. Described is what the source code leads one to expect.

1. Export: in the UI "Colors & filaments" → **3MF multi-color** (with parts, one object per part) or `cad export modell.scad -o modell.3mf`. The color list shows slot, volume and mass per color beforehand; slots are changed via the selector (writes `filament = n`).
2. In Bambu Studio choose **File → Import → 3MF/STL... (Ctrl+I)** (or drag the file into the window, or "Open project"). The file does not come from Bambu Studio; expect the notice that only geometry (and colors) are loaded – that is correct.

3. **Color dialog** (current versions): Bambu Studio shows the colors of the file (e.g. white, teal, red, blue, yellow) and suggests filaments for them. **"Append"** (German UI: „Anhängen“) creates the colors as new filaments; mapping to existing filaments puts the colors on the AMS slots; confirm. With colors per sub-part, each sub-part stays entirely in its filament.
4. Without the color dialog (older version, OrcaSlicer or cancelled): the slots from the file apply. In the **object list** (left) the objects `schild` and `figur` appear; expanded, one sub-part per color ("schild – white", "schild – teal", "schild – red" ...), next to it the filament number (1, 2, 3 ...). Bambu Studio adds missing filaments on import up to the highest slot (with its own colors); then choose the right colors/AMS slots in the filament bar ("Sync AMS").
5. Check: the sub-parts of an object sit exactly on each other (common origin), the lettering stands as relief on the plate, the collar sits at the neck of the figure. "Arrange objects" only moves whole objects.
6. **Slice**: in the preview choose the view "Filament" – each color in its color, filament changes at the color boundaries, prime tower with several filaments. Compare the mass per filament with the color list (PLA 1.24 g/cm³; the slicer computes with walls/infill, hence different).

OrcaSlicer: as above without the color dialog – sub-parts and slots from `model_settings.config`, set the filament colors yourself. PrusaSlicer does not read this metadata: there use "STL per color (ZIP)", load all STLs together and answer **Yes** to "Multi-part object?", then set the extruder per sub-part (file names `F<slot>_<farbe>.stl`).

11 Known limitations

- Color regions depend on SMLib's booleans: where SMLib loses material (e.g. union with a torus, known since report 1 – the example therefore uses a ring made from a cylinder), the regions are wrong; the engine warns ("sum ≠ volume") or reports an error. About three booleans per color node; the UI requests the regions only while the color list is open.
- Bodies that meet only along a line do not give a closed body (found with the first letter "D": two bars touched only along an edge, lib3mf reported the mesh as non-manifold; now they overlap).
- `color()` in 2D subtrees (below `linear_extrude` etc.) colors faces but does not split volume. Material outside the parts is missing from the color list if there are parts.
- Filament colors in Bambu Studio via the color dialog or by hand (no `project_settings.config`, see 5.6); not checked in Bambu Studio and OrcaSlicer in the container. PrusaSlicer: components become separate objects, no slots.
- Color per face in the view via the face tag: two bodies with identical content and different colors share one tag and both show the first color (known limitation of the provenance); "Show color regions" is not affected.
- Color vision deficiency: see section 2 (smallest ΔE with deuteranopia 0.070, tritanopia 0.031). White on a light and black on a dark background distinguishable only by shading and edges.
- Theme "System" in Tauri depends on the GTK color scheme; under Xvfb without a scheme it is light. Measurements and images under Xvfb/SwiftShader without GPU.

12 Reproduction

```
export SMLIB_PREBUILT=$PWD/smlib/build-native CAD_NO_DAEMON=1
cargo build --release -p cad-cli                  # nach: cd web && pnpm build:tauri
cargo test --workspace --no-fail-fast > reports/tmp/report11/cargo.txt 2>&1
cargo clippy --workspace --all-targets -- -D warnings > reports/tmp/report11/clippy.txt 2>&1
cargo fmt --all --check > reports/tmp/report11/fmt.txt
(cd web && pnpm typecheck && pnpm test > ../reports/tmp/report11/vitest.txt && pnpm e2e)
cp reports/tmp/report2/playwright.json reports/tmp/report11/
node web/e2e-tauri/desktop.e2e.mjs --bin target/release/cad --out reports/tmp/report11/tauri
CAD=target/release/cad python3 tests/openscad-compat/compare.py --eval > reports/tmp/report11/compare.txt
/opt/tools/venv-step/bin/python tests/step/verify.py --cad target/release/cad > reports/tmp/report11/step.txt
CAD=target/release/cad PY3MF=/pfad/venv/bin/python tools/report/report11-data.sh   # lib3mf, trimesh
node tools/report/report11.mjs
node tools/report/print-pdf.mjs docs/reports/11-farben-filamente/report.html docs/reports/11-farben-
filamente/report.pdf "Test report 11 – colors, multi-filament 3MF, dark theme"
```